Guided Filtering of ARFI Displacement Data for Shear Wave Speed Estimation

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***Abstract*—Acoustic radiation force impulse (ARFI) imaging enables tissue stiffness characterization by inducing shear waves with a push pulse and subsequently tracking the resulting tissue displacements using plane-wave pulses. Such displacement data can then be utilized to get a 2D map of the estimated shear wave speed (SWS) values within an imaged region of interest. This is a challenging task because displacement estimates are typically noisy. To mitigate the impact of noise, we propose to apply guided image filtering that relies on two types of displacement data: one based on progressive referencing (i.e., using correlation between consecutive beamformed frames), and the other based on fixed referencing (i.e., using correlation with respect to some fixed pre-push frame). The latter serves as the source of guidance images that help denoise the progressive-referencing data prior to SWS estimation. This technique was applied to the ARFI data of the IEEE UFFC-JS 2024 a-MEM Challenge.**

***Index Terms*—shear wave elastography, acoustic radiation force, tissue displacement, ultrafast imaging, guided filtering**

I. Introduction

Acoustic radiation force impulse (ARFI) imaging is a widely used method for assessing biomechanical properties of soft tissues [1]. It involves three basic steps: 1) inducing shear waves with a push pulse, 2) tracking the resulting tissue displacements using plane-wave pulses, and 3) processing acquired data to estimate a 2D shear wave speed (SWS) map within an imaged region of interest. Such a map provides important diagnostic information about tissue elasticity.

The SWS mapping process consists of the following computational tasks [1]: 1) estimation of tissue displacements from a sequence of beamformed image frames (e.g., see [1]–[3]), 2) filtering of displacement data (e.g., see [4]), and 3) estimation of the local SWS values (e.g., see [5]). We focus on the second task and propose to apply *guided image filtering* [6] as an additional mechanism for displacement data denoising.

The main idea is to employ two types of displacement data: one based on *progressive referencing* (i.e., using correlation between consecutive beamformed frames), and the other based on *fixed referencing* (i.e., using correlation with respect to some fixed pre-push frame) [1]. We utilize the latter to extract guidance images for filtering the progressive-referencing data prior to SWS estimation.

Section II briefly describes guided filtering details in the context of ARFI imaging. Section III visually illustrates the effect of applying the proposed method to the tissue-mimicking phantom data provided by the organizers of the IEEE UFFC-JS 2024 a-MEM Challenge [7]. Our concluding summary is given in Section IV.

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II. Proposed Approach

Let $A(z,x,t)$ and $B(z,x,t)$ denote 3D axial displacement data volumes that correspond to a temporal sequence of 2D frames computed using the progressive and fixed referencing schemes, respectively. Symbol t refers to time, while the axial and lateral point coordinates on the 2D image grid are given by (z,x). In the sequel, when z is fixed, the corresponding individual 2D frames are denoted by $A(x,t)_z$ and $B(x,t)_z$. Similarly, when we fix x or t, the individual 2D frames are denoted by $A(z,t)_x$ and $B(z,t)_x$, or by $A(z,x)_t$ and $B(z,x)_t$, respectively. Our goal is to estimate a 2D SWS map $\hat{v}(z,x)$ from the displacement data.

First, given the complex-valued in-phase/quadrature beamformed data $\mathrm{IQ}(z,x,t)$, we estimate displacements $\hat{u}_A$ and $\hat{u}_B$ using equations (1) and (2) that can be found in references [2] and [1]. Symbols c and f_c represent the speed of sound (used during beamforming) and the center frequency of the RF data before demodulation. Symbols I and Q refer to the in-phase and quadrature components of the selected IQ data samples. In (1), the sample selection is performed by means of an $M \times L$ sliding window moving across 2D data frames $A(z,t)_x$ and $B(z,t)_x$ for every fixed x. Symbols m and l designate axial and temporal indices within the sliding window. On the other hand, indices 0 and j in (2) represent fixed time instances t_0 (pre-push reference) and t_j (current beamformed data).

Second, for every fixed t, we apply a 2D median filter of size 1×1 mm^2 to $A(z,x)_t$ and $B(z,x)_t$ computed in the previous step. Both $A(z,x)_t$ and $B(z,x)_t$ are then normalized to their respective maximum absolute data values.

Next, for every depth z, we use a 2D directional filter from [4] to remove potential reflection artifacts. This is followed by 2D median filtering with the neighborhood size of 1×1 mm$\cdot$ ms, applied to both $A(x,t)_z$ and $B(x,t)_z$. Finally, as a novel step proposed here, we perform guided filtering of $A(x,t)_z$ utilizing $B(x,t)_z$ as the guidance image. The resulting data is denoted by $A'(x,t)_z$. By repeating this process for every fixed z, we obtain $A'(z,x,t)$ used for the subsequent calculations of $\hat{v}(z,x)$ (see Section III).

$$\hat{u}_A = \frac{c}{4\pi f_c} \frac{\arctan\left(\frac{\sum_{m=0}^{M-1}\sum_{l=0}^{L-2}[Q(m,l)I(m,l+1)-I(m,l)Q(m,l+1)]}{\sum_{m=0}^{M-1}\sum_{l=0}^{L-2}[I(m,l)I(m,l+1)+Q(m,l)Q(m,l+1)]}\right)}{1+\arctan\left(\frac{\sum_{m=0}^{M-2}\sum_{l=0}^{L-1}[Q(m,l)I(m+1,l)-I(m,l)Q(m+1,l)]}{\sum_{m=0}^{M-2}\sum_{l=0}^{L-1}[I(m,l)I(m+1,l)+Q(m,l)Q(m+1,l)]}\right)/2\pi} \tag{1}$$

$$\hat{u}_B = \frac{c}{4\pi f_c} \arctan\left(\frac{\sum_{m=0}^{M-1}Q(m,0)\sum_{m=0}^{M-1}I(m,j) - \sum_{m=0}^{M-1}I(m,0)\sum_{m=0}^{M-1}Q(m,j)}{\sum_{m=0}^{M-1}I(m,0)\sum_{m=0}^{M-1}I(m,j) + \sum_{m=0}^{M-1}Q(m,0)\sum_{m=0}^{M-1}Q(m,j)}\right) \tag{2}$$

$$C_n(q) = \frac{\sum_p \left[D(n-w/2,p)-\bar{D}(n-w/2)\right]\left[D(n+w/2,p+q)-\bar{D}(n+w/2)\right]}{\sqrt{\sum_p \left[D(n-w/2,p)-\bar{D}(n-w/2)\right]^2 \sum_p \left[D(n+w/2,p+q)-\bar{D}(n+w/2)\right]^2}} \tag{3}$$

In the context of ARFI displacement data processing, the concept of a guided image filter [6] can be explained as follows. Let ω_k represent a 2D rectangular window centered at some pixel location k. Assuming that another pixel location i is contained within ω_k, let A_i and B_i represent individual data samples from $A(x,t)_z$ and $B(x,t)_z$ whose (x,t) coordinates represent location i in question. The guided filter relies on the following linear relationship:

$$A'_{i,k} = \alpha_k B_i + \beta_k, \quad \forall i \in \omega_k, \tag{4}$$

where $A'_{i,k}$ is the output pixel value specific to ω_k. The filter coefficients are given by

$$\alpha_k = \frac{\frac{1}{|\omega_k|}\sum_{i\in\omega_k} A_i B_i - \bar{A}_k \bar{B}_k}{\sigma_k^2 + \epsilon}, \quad \beta_k = \bar{A}_k - \alpha_k \bar{B}_k, \tag{5}$$

where σ_k^2 is the variance of the guidance image pixels within ω_k, while $\bar{A}_k$ and $\bar{B}_k$ are the mean values of the input and guidance image pixels within ω_k, respectively. The above coefficients arise from the optimal solution to the minimization problem associated with the cost function

$$\text{COST}(a_k, b_k) = \sum_{i\in\omega_k} \left[(\alpha_k B_i + \beta_k - A_i)^2 + \epsilon\alpha_k^2\right], \tag{6}$$

where ϵ is the regularization parameter. The final output of the guided filter is

$$A'_i = \frac{1}{N_\omega} \sum_{k|i\in\omega_k} A'_{i,k}, \tag{7}$$

which is an average over all N_ω overlapping windows ω_k that contain pixels A_i and B_i.

III. Illustrative Results

The proposed method was applied to the experimental ARFI data of the IEEE UFFC-JS 2024 a-MEM Challenge [7]. Three tissue-mimicking phantoms were imaged using two pushes (induced near the left and right boundaries of the insonified region of interest), and the resulting displacements were tracked using repeated triples of plane-wave pulses (emitted at three steering angles $-4°$, $0°$, and $4°$).

Beamforming with coherent compounding produced 110 IQ data frames (5 pre-push and 105 tracking frames) spaced at the regular time intervals of $\Delta t = 285$ μs. The frame size was 325×247, with the axial and lateral ranges of $0 \le z \le 50.204$ mm and $-18.961 \le x \le 18.961$ mm, respectively. The grid resolution was set to $\Delta z = \Delta x = 0.1540$ mm. Fig. 1 displays B-mode images of the two phantoms with inclusions, whose approximate boundaries are marked in red. The corresponding SWS maps are displayed in Fig. 2.[1] The third (homogeneous) phantom is not shown, as its visual appearance is similar to the background of the left-hand side image in Fig. 2.

We have used normalized cross-correlation [2] to estimate the local SWS values as follows. For each depth z, we extract the corresponding 2D data slice $A'(x,t)_z$ and then employ a sliding window of size $w\Delta x$ moving along the x-axis. For a given lateral position index n at the center of the window, we treat the displacement traces at indices $n-w/2$ and $n+w/2$ as the reference and target signals, respectively. They are denoted by $D(n-w/w,p)$ and $D(n+w/w,p)$, where p refers to the temporal index within that window.[2] Their normalized cross-correlation $C_n(q)$ is calculated as shown in (3), where q is the time lag applied to align the two signals in question. Symbols $\bar{D}(n-w/2)$ and $\bar{D}(n+w/2)$ in (3) represent the respective mean values of the reference and target signals over their temporal span. The shear wave propagation delay is found by identifying the optimal time lag q that maximizes the value of

[1] It should be mentioned that during SWS mapping, the range of allowable values was limited to $[0.2, 5.0]$ m/s for the phantom on the left and $[0, 5.0]$ m/s for the phantom on the right (see Fig. 2 and Fig.3).

[2] Here, the temporal span of our sliding window included all displacement tracking frames.

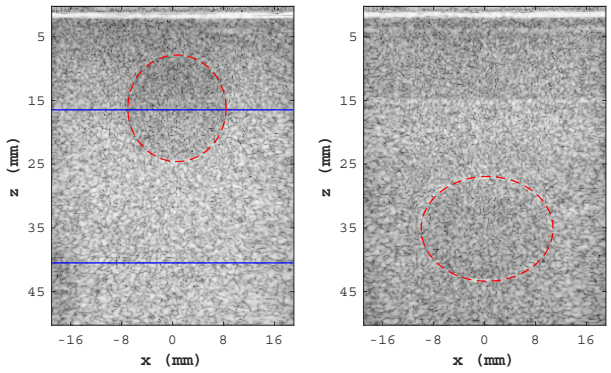


Fig. 1. B-mode images of two phantoms with inclusions (60-dB range).

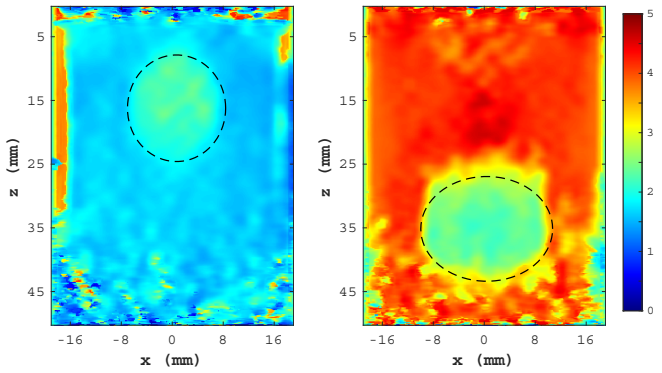


Fig. 2. SWS maps with guided filtering applied to displacement data.

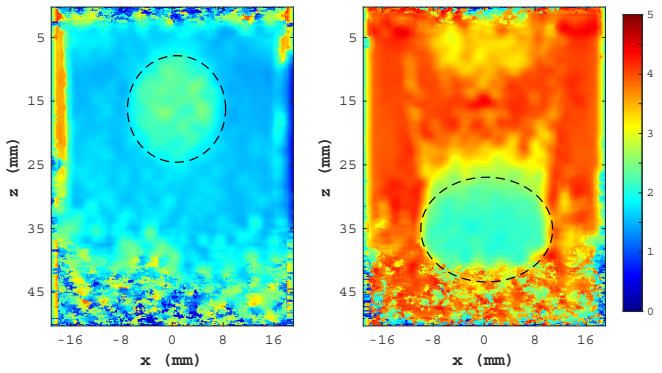


Fig. 3. SWS maps without guided filtering of displacement data.

$C_n(q)$. That is,

$$\tau = \Delta t \cdot \underset{q}{\operatorname{argmax}}\, C_n(q). \tag{8}$$

The SWS estimate at the corresponding location (z, x) is set to $\hat{v} = w\Delta x/\tau$. In this work, we used $w = 13$ corresponding to the window size of 2 mm. After obtaining the SWS values for all grid points of interest, we smooth out the final 2D map by applying a simple edge-preserving 2D Wiener filter of size 2×2 mm^2.

As mentioned earlier, the resulting SWS maps for the two phantoms with inclusions are displayed in Fig. 2. Our SWS mapping process was somewhat successful, as indicated by the expected inclusion boundaries drawn on the same images (see also Fig. 1). For comparison, Fig. 3 displays the SWS maps obtained from the displacement data without guided filtering. Note that the bottom section of the left-hand side image is more noisy, while the inclusion shape in right-hand side image is less accurate.

To further illustrate the effect of guided filtering, Fig. 4 and 5 show two examples of the progressive-referencing data slices $A(x,t)_z$ and $A'(x,t)_z$, i.e., before and after guided filtering, along with their respective guidance images. They correspond to the depths z approximately equal to 16.5 and 40.5 mm, marked in blue on the left-hand side image in Fig. 1. As one can see in Fig. 4, guided filtering effectively performed edge-preserving smoothing. In Fig. 5, it also transferred structural features of the guidance image, thus enhancing the prominence of shear wave peaks. Additionally, for the same phantom, Fig. 6 shows several displacement data frames corresponding to the time instances t = 2.56, 7.69, 12.82, 17.95, and 23.08 ms. It provides another illustration of the denoising effect of guided filtering as described in this work.

IV. Conclusion

Focusing on the filtering step of the SWS mapping process, we have proposed a novel additional step for denoising ARFI displacement data, based on the concept of a guided image filter [6]. Specifically, we employ the fixed-referencing data as the source of guidance images for filtering the progressive-referencing data prior to SWS estimation. We have applied our method to the experimental data of the IEEE UFFC-JS 2024 a-MEM Challenge, visually demonstrating potential benefits of the proposed approach. It should be noted that our guided filtering can be used in conjunction with other SWS estimation methods (e.g., see [8], [9]) that are likely to give more accurate SWS maps compared to ours.

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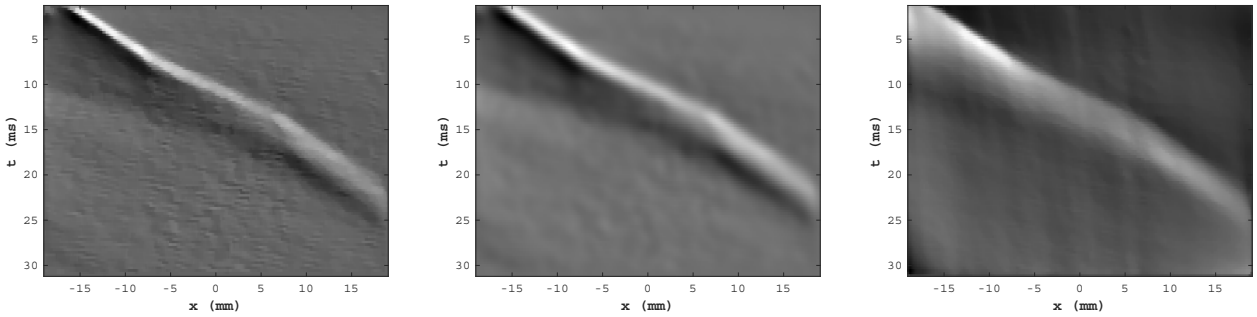


Fig. 4. Data slice at depth $z = 16.5$ mm, before (left) and after (center) guided filtering, using a given guidance image (right).

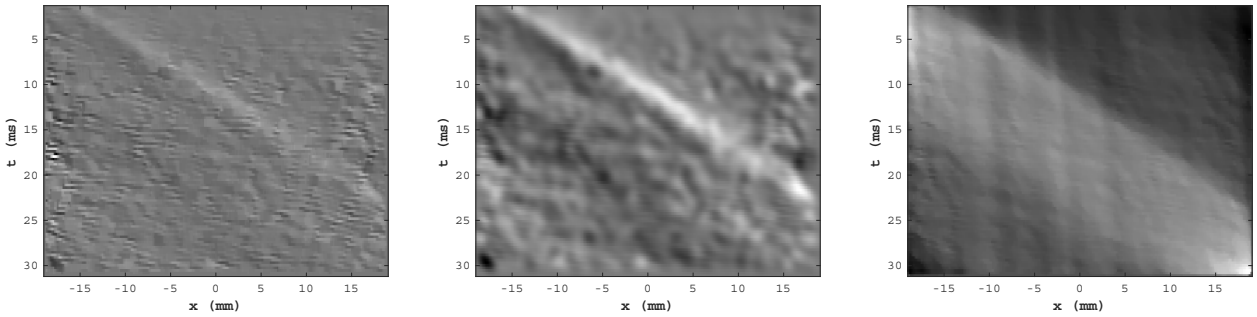


Fig. 5. Data slice at depth $z = 40.5$ mm, before (left) and after (center) guided filtering, using a given guidance image (right).

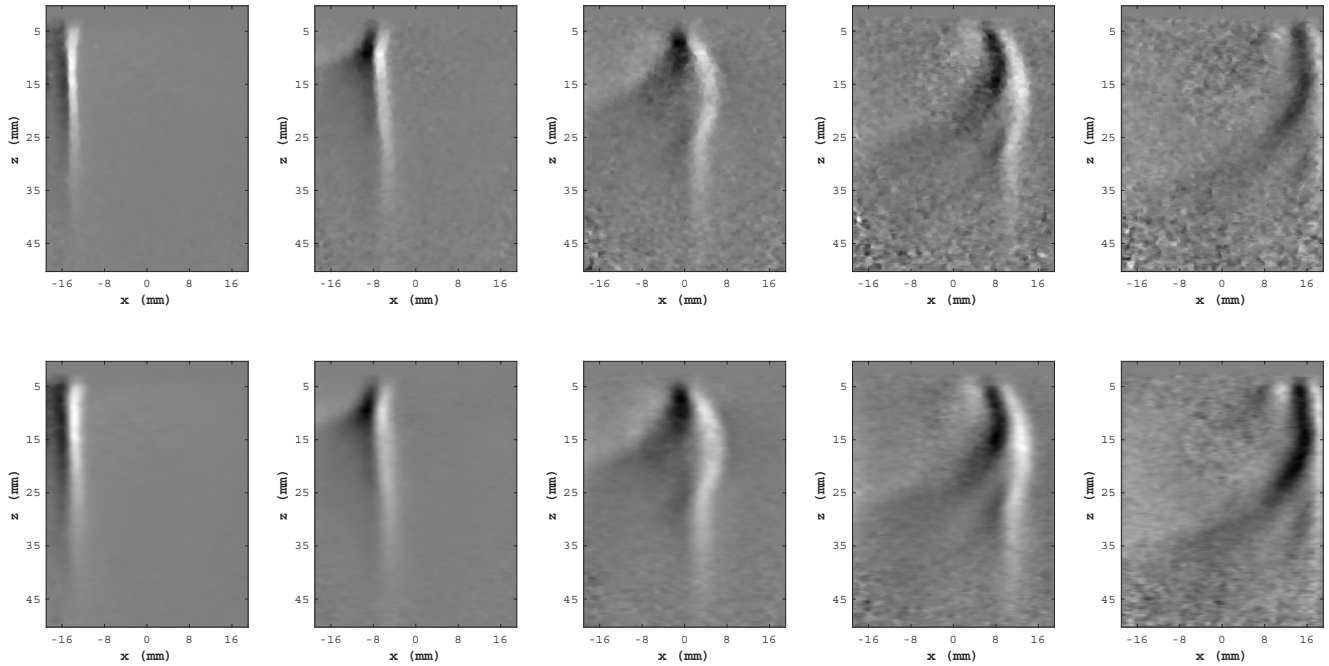


Fig. 6. From left to right: Data frames at $t = 2.56, 7.69, 12.82, 17.95, 23.08$ ms, before (top) and after (bottom) guided filtering.